

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
WORK INTEGRATED LEARNING PROGRAMMES (WILP) DIVISION
COURSE: SE* ZC343 / SE* ZC461 / IS ZC343 – SOFTWARE ENGINEERING

Submission Date: 9th August 2026

Assignment Submission Mode: ELearn LMS- File Upload

Weightage: 20 %

1. OBJECTIVE:

The objective is to Translate real-world business challenges into structured Software Requirements Specifications (SRS) using industry-standard templates. And then Develop behavioral and structural analysis models (Scenario, Flow, and Class modeling) to map out system behavior. Apply software architectural principles to design a scalable system using the 4+1 view model. And also formulate algorithmic or empirical effort estimation models to predict project resource and schedule requirements.

This is an individual assignment to be carried out independently.

2. PROBLEM STATEMENT

TITLE: 'ECODRIVE' SMART CAMPUS EV CHARGING & GRID MANAGEMENT SYSTEM

Context: Your university campus wants to transition to a green campus by deploying an automated Electric Vehicle (EV) Charging & Grid Management System (EcoDrive). The campus will feature 50 smart EV charging stations servicing faculty, students, and campus utility vehicles.

System Challenges & Core Features:

- **Intelligent Slot Allocation:** Drivers must be able to book charging slots via a mobile app. The system must prioritize allocation based on user roles (e.g., emergency vehicles > faculty > students) and state-of-charge (SoC) data sent by the vehicle.
- **Dynamic Load Balancing:** The university campus has a capped power grid limit. If total power demand exceeds 85% capacity, the system must dynamically throttle power distribution to low-priority vehicles.
- **Billing & Analytics Integration:** Integrated payment processing for electricity usage with real-time dashboards for campus administrators to track power consumption and revenue.

Your team has been tasked as the software engineering division to take this raw conceptual requirement through the core planning and architectural design phases.

3. ASSIGNMENT DELIVERABLES & EVALUATION RUBRIC

Task 1: Software Requirements Specification (SRS) [5 Marks]

- Draft a focused SRS document for the EcoDrive system adhering to IEEE 830 standards.
- Functional Requirements: Detail at least 5 core functional requirements covering booking, load balancing, and billing.
- Non-Functional Requirements (NFRs): Specify at least 3 critical NFRs (e.g., availability under peak load, security of payment data, response time for load throttling) with measurable metrics.

Task 2: Requirement Analysis & Behavioral Modeling [5 Marks]

- Construct the analysis model artifacts to visually demonstrate how data and control flow through the ecosystem:
- Scenario-Based Modeling: Draw a comprehensive Use Case Diagram representing all key actors (Drivers, Admin, Smart Grid API).
- Flow & Behavioral Modeling: Create an Activity Diagram mapping the entire lifecycle of a user arriving, authenticating, charging, and checking out, explicitly showing the dynamic load balancing decision logic.

Task 3: Architectural Design (4+1 View Model) [5 Marks]

- Architect a scalable solution to handle thousands of concurrent state-of-charge updates.
- Logical View: Design a detailed Class Diagram showcasing the structural components (e.g., User, Vehicle, ChargingSlot, LoadController, Transaction) and their relationships (associations, aggregations).
- Development/Implementation View: Propose a suitable architectural style (e.g., Microservices, Event-Driven, or Layered). Justify why your selected style perfectly matches the system's scalability and real-time throttling needs.

Task 4: Project Estimation and Sizing [5 Marks]

- Quantify the effort required to build the EcoDrive MVP (Minimum Viable Product).
- Calculate the total Function Points (FP) or Use Case Points (UCP) for the features outlined in your SRS.
- State your environmental and complexity assumptions clearly (e.g., team experience, technical complexity factors).
- Compute the estimated Effort (Person-Months) and Development Time using basic COCOMO or an equivalent empirical model.

4. TOOLS AND TECHNOLOGIES REQUIRED

- Requirements Documentation: Microsoft Word, Google Docs, or LaTeX equipped with an IEEE 830 SRS Template.

- System Modeling and Diagrams: Draw.io (diagrams.net), Lucidchart (collaborative), StarUML (desktop modeler), or PlantUML (text-to-diagram).
- Metrics and Estimation Calculators: Microsoft Excel or Google Sheets (pre-configured with formulas)
- or open-source online COCOMO II Calculators.

5. AI USAGE POLICY AND REFLECTION REPORT:

Because this assignment measures foundational critical thinking, GenAI tools should be treated as collaborative brainstorming assistants rather than automated completion tools.

Permitted Use Cases (Green Light):

- Brainstorming Edge Cases: Identifying potential edge cases or operational risks unique to smart grid APIs.
- Syntax and Formatting: Formatting text into structured Markdown or debugging PlantUML script syntax.
- Understanding Formulas: Asking AI to explain the underlying mechanisms of COCOMO scale factors or Function Point formulas.

Prohibited Use Cases (Red Light):

- Direct Prompting for Deliverables: Copy-pasting the assignment prompt to completely generate text or structural layout outlines.
- AI-Generated Diagrams: Exporting charts generated completely by specialized AI tools without manual customization.
- Blind Calculations: Turning in estimation figures without showing your step-by-step mathematical reasoning.

Mandatory AI Disclosure Statement:

If Generative AI is utilized in this assignment, students must append an AI Disclosure Appendix at the very end of their report listing: (1) Tool name, (2) Exact prompts used, and (3) A 2-sentence Description validation detailing how outputs were critically verified.

6. SUBMISSION INSTRUCTIONS

1. Format: Submissions must be uploaded as a single PDF document in the taxilla coursepage only. No email submissions will be accepted.
2. Tooling: Use open-source or academic versions of modeling tools (e.g., draw.io, StarUML, Lucidchart) for diagrams.



3. Academic Integrity: Plagiarism in diagrams or requirements text will result in a zero score. Originality in architectural justification and estimation assumptions is heavily weighted. You can discuss with peers and refer to the internet to understand the concept better. However, you may not share your work with other students or do not do a verbatim copy from the Generative AI/Internet/GitHub repositories. Include appropriate references. The code will be tested for plagiarism. If found guilty, no marks will be awarded.

7. EVALUATION RUBRIC:

Total: 20 Marks

Activity	Marks
Task 1: Software Requirements Specification (SRS)	5
Task 2: Requirement Analysis & Behavioral Modeling	5
Task 3: Architectural Design (4+1 View Model)	5
Task 4: Project Estimation and Sizing	5

The relevant documents to be submitted only in the elearn portal. After the completion of deadline, Assignment will be evaluated by a group of instructors on the submissions made.

NOTE:

- This is a take-home assignment to be carried out by each student independently
- In case of any further queries, use discussion forums, or reach out to me at akshaya.ganesan@pilani.bits-pilani.ac.in.